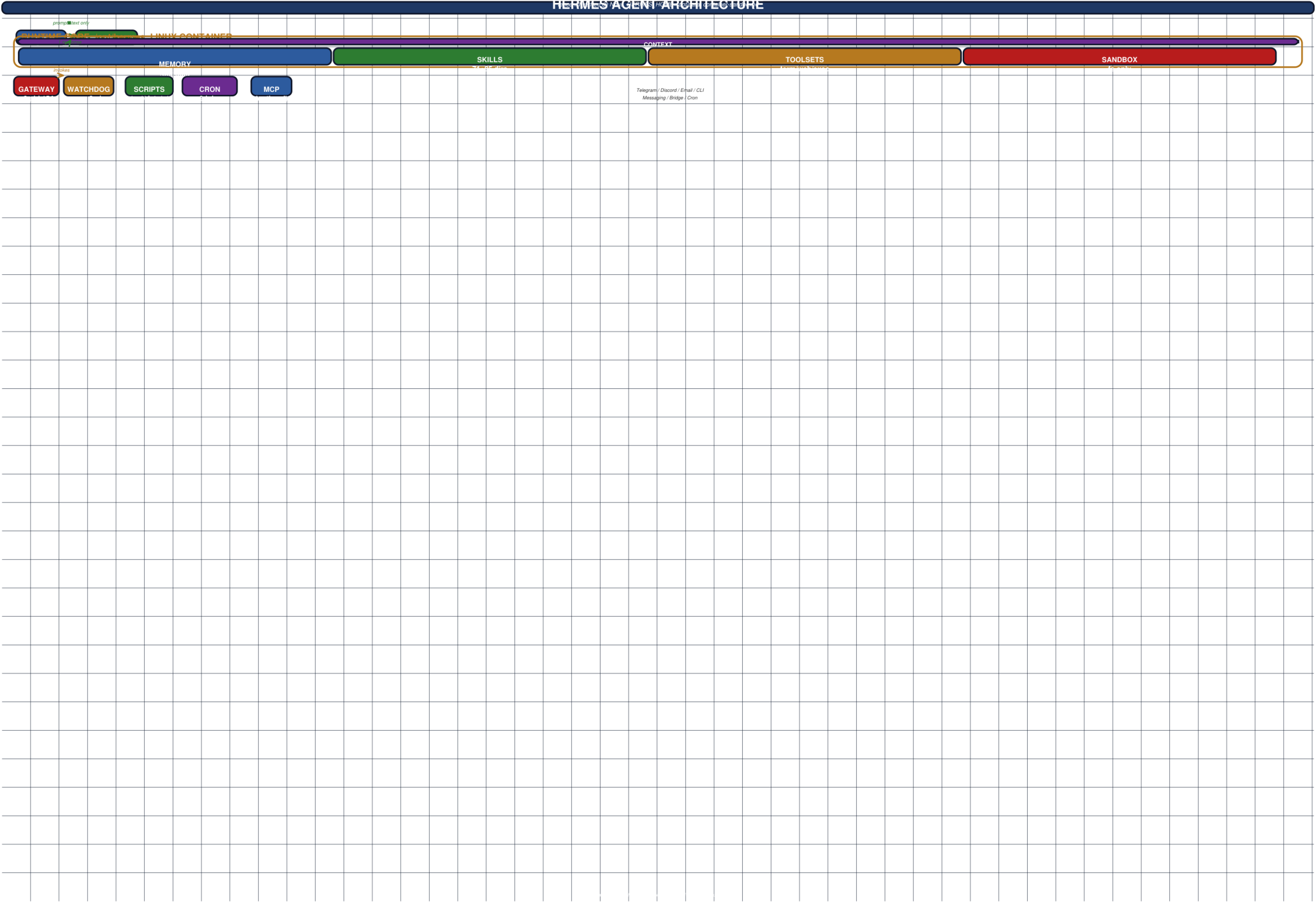




Context Pipeline

The active context is rebuilt every turn and sent as ONE request to the single pinned model:

Step	Source	Type	Persists
1	System prompt (bundled)	static	no
2	MEMORY.md + USER.md	auto-injected	yes - disk file
3	Skills INDEX (names only)	flat list	no
4	Conversation history (this session)	compactable	no
5	New user message	input	no

History compaction: oldest verbatim turns are summarized+dropped when context grows; gist retained. Across sessions, history resets — only MEMORY.md/USER.md and state.db persist (via session_search).

Model Selection & Skills

Two misreadings: (1) NO free-tier model router in default agent, (2) 81 skills = 71+85 deduped.

Aspect	Reality vs Misconception	
Model routing	Profile config — ONE active model per session	(per-task route
Active model	tencent/hy3:free @ Nous (pinned)	(multi-model a
Free-tier	NOT in default agent — was a design spec only	(auto-routed p
Paid	Explicit UI approval only	(autonomous

Skill count: 71 bundled (/opt/hermes/skills) + 85 profile (/opt/data/skills, incl .hub) = 81 deduped.

Tool Execution & Sandboxing

All tool execution inside the Linux container. Nous sees only text of calls+results.

Toolset	Execution environment	Boundary to Nous
terminal	Shell + python3 in Linux container	Container fs only
execute_code	Script in session venv	Persistent venv, same container
file (read/write/patch)	Direct runtime OS ops	Path lock + staleness
web/image/skills	FAL / search APIs	Same text channel

Limits: 300s timeout, 50 tool calls/turn, 50KB stdout, 10KB stderr. write_file lint gate fail-closed: .py->py_compile, .js->node --check, .json/.yaml/.toml->parse, .ts/.go/.rs->shell linter. verified:true = hash confirmed.

Cron Daemon & Delivery Layer

Cron Daemon (9 jobs) runs via s6 supervision under the Messaging Gateway.

Job type	Jobs	How it runs	Output delivered
no_agent	watchdog, git-push, cleanup, catchup	Script only — stdout verbatim	Silent = success
agent	mentor, pensolar, coo, marketing	LLM reasoning on prompt	Reasoned briefing

Path rule: no_agent scripts must live under /opt/data/scripts/. repeat=-1 = infinite. startup-catchup armed + forever.

Memory, State & Persistence

Three distinct persistence mechanisms — do not conflate them:

Mechanism	What it is	Scope	How accessed
MEMORY.md / USER.md	Durable fact files	All sessions	Auto-injected every turn
state.db (SQLite)	Session store + facts DB	All sessions	session_search
Conversation history	This session messages	Current session only	In-context; compacted when long

The model receives: system prompt + MEMORY.md + USER.md + skills INDEX + conversational history + new message. It sees neither the filesystem nor internal state — only this flattened text.

Security Model & Data Flow

The container is a hard boundary. Nous is outside and sees only text.

Trust zone	Inside container	Outside container
Container (/opt/data, /opt/hermes)	Tools, files, venvs, state.db	- (Nous)
Nous provider channel	Flattened context text only	Filesystem/disk
Browser (Windows host)	Read localhost:8765 for doc-reader	Cannot reach internals
Browser-use daemon	Headless Chromium (sanitized)	Sandboxed: no file:// or localhost

Data never crosses the boundary in binary form — all tool results are serialized to text before reaching the model.